

Prognostic value of tumor size varies by treatment in a meta-analysis of 15 randomized clinical trials in advanced non-small cell lung cancer across immunotherapy, TKI, and chemotherapy regimens

Jacqueline Buross¹, Krzysztof Sakrejda¹, Daniel Lee¹, Eric Novik¹, and Philip Jewsbury²

(1) Generable Inc, and (2) AstraZeneca Plc

Introduction

In the context of oncological drug development, we need well-calibrated, reliable, early indicators of drug efficacy in order to allocate research money and effort wisely.

RECIST [1] is commonly used to characterize intermediate outcomes for clinical trials in the context of solid tumors. It is based largely on a standard measure of tumor size known as the sum of longest diameters (SLD).

Here, we evaluate whether the prognostic value of tumor size varies depending on the class of treatment.

Population

Data for 6,156 subjects from 15 randomized clinical trials in non-small-cell lung cancer were included in this analysis, spanning treatments from targeted, immunotherapy, and cytotoxic treatment classes as both first- and second-line therapies. Several arms evaluated combination therapies. Two of the studies were sourced from project datasphere [7], and so those studies only include subjects in the control arms. These data and 13 other studies were provided by AstraZeneca under a collaboration agreement.

Trial	Cohort	Treatment	N
INTEREST (NCT0076388, 2+ line, 2004-2007)	Stage III-IV NSCLC	docetaxel	396
		gefitinib	413
IPASS (NCT00322452, first line, 2006-2008)		carboplatin + paclitaxel	375
		gefitinib	402
ZODIAC (NCT00312377, 2+ line, 2006-2008)	Stage IIIb-IV NSCLC	docetaxel	538
ABRAXANE-CA031 (NCT00540514, first line, 2007-2009)		carboplatin + taxol	463
IPUM (NCT0126917, first line, 2010-2012)		gefitinib	84
IMPRESS (NCT01544170, 2+ line, 2012-2014)		gefitinib + pemetrexed	51
	EGFR-mutated (activating)	pemetrexed	48
	PD-L1 low/negative (<25%)	durvalumab	94
	PD-L1 positive (>=25%)	durvalumab	107
	PD-L1 unknown	durvalumab	16
Study 1108 (NCT01093562, first line, 2013-2019)		docetaxel	2
	KRAS unknown	docetaxel + selumetinib	11
	KRAS wild type	docetaxel	23
		docetaxel + selumetinib	92
SELECT-2 (NCT01750281, second line, 2012-2016)		docetaxel	7
		docetaxel + selumetinib	22
	KRAS-mutated	docetaxel	224
		docetaxel + selumetinib	209
SELECT-1 (NCT01933032, second line, 2013-2016)		durvalumab	220
	EGFR+/ALK-	durvalumab	61
	EGFR-/ALK- with PD-L1 >= 90%	durvalumab	90
	EGFR+/ALK+	durvalumab	189
PACIFIC (NCT01254611, 3+ line, 2014-2017)	Stage III NSCLC	chemoradiation	389
		carboplatin + cisplatin + pemetrexed	44
AURA3 (NCT02151981, 2+ line, 2014-2016)	EGFR-mutated (T790M)	osimertinib	108
		durvalumab	108
	PD-L1 low/negative (<25%)	durvalumab + tremelimumab	162
	standard of care	tremelimumab	56
ARCTIC (NCT0252948, 2+ line, 2015-2018)		durvalumab	61
	PD-L1 positive (>=25%)	standard of care	54
CAURAL (NCT02454933, first line, 2015-2017)	EGFR-mutated (T790M)	durvalumab + osimertinib	11
		osimertinib	15
	PD-L1 low/negative (<25%)	durvalumab	170
	standard of care	durvalumab + tremelimumab	177
		standard of care	162
MYSTIC (NCT02452882, first line, 2015-2018)		durvalumab	136
	PD-L1 positive (>=25%)	durvalumab + tremelimumab	133
	standard of care		111
Total			6156

Key Findings:

- As expected, there is a strong association between change in tumor size (both initial shrinkage and eventual regrowth) and overall survival.
- Among patients with the same degree of tumor shrinkage, those on a PD-1/PD-L1 inhibitor have a lower rate of mortality than those receiving an EGFR inhibitor, MEK inhibitor, CTLA-4 inhibitor, or chemotherapy regimen.

Model Description

The model is a hierarchical meta-analytic joint model for longitudinal tumor size and overall survival. It is estimated using Bayesian inference in Stan [6].

It contains a sub-model for tumor size (TS) over time [5]:

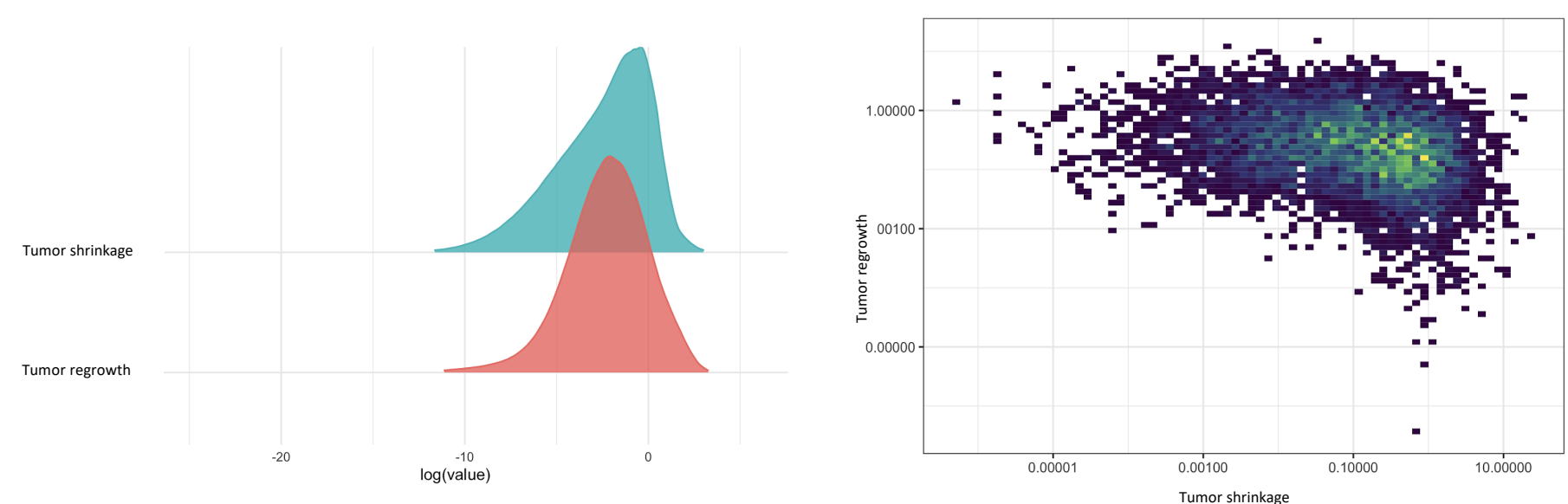
$$TS(t) = TS_0 \times [(f) \exp(-k_s t) + (1 - f) \exp(k_g t)]$$

where parameters k_s and k_g govern the rate of shrinkage and growth among susceptible (f) and resistant (1-f) cells, respectively. These are estimated per patient and per class of drug using component GLM sub-models.

We characterize each patient's tumor-size trajectory using two quantities:

$$\text{shrinkage}_i = f_i \cdot k_{s,i} \cdot 100$$
$$\text{regrowth}_i = (1 - f_i) \cdot k_{g,i} \cdot 1000$$

The quantities are rescaled so that they have similar median values and inter-quartile ranges.



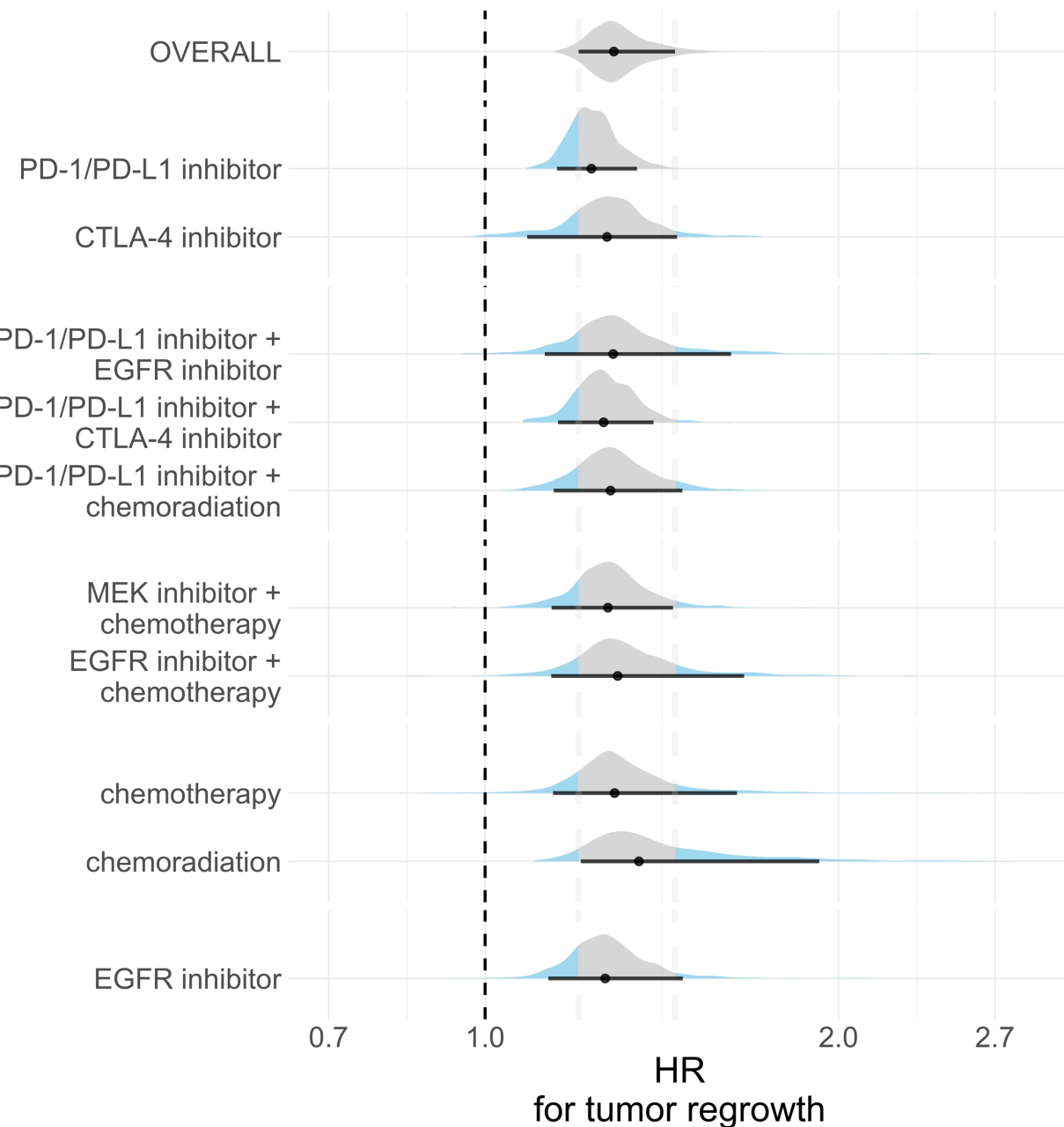
These two quantities are then included as covariates in a Cox PH meta-analytic survival model with stratified baseline hazard (by study & cohort) and covariate effects that vary by trial arm and treatment class.

$$h_i(t) = h_0(t, \omega) \times \exp(\beta_1 \text{shrinkage}_i + \beta_2 \text{regrowth}_i)$$

The TS and hazard models are estimated jointly so that the full uncertainty in the estimates of shrinkage and regrowth are included in the hazard model.

Effect of tumor regrowth on survival

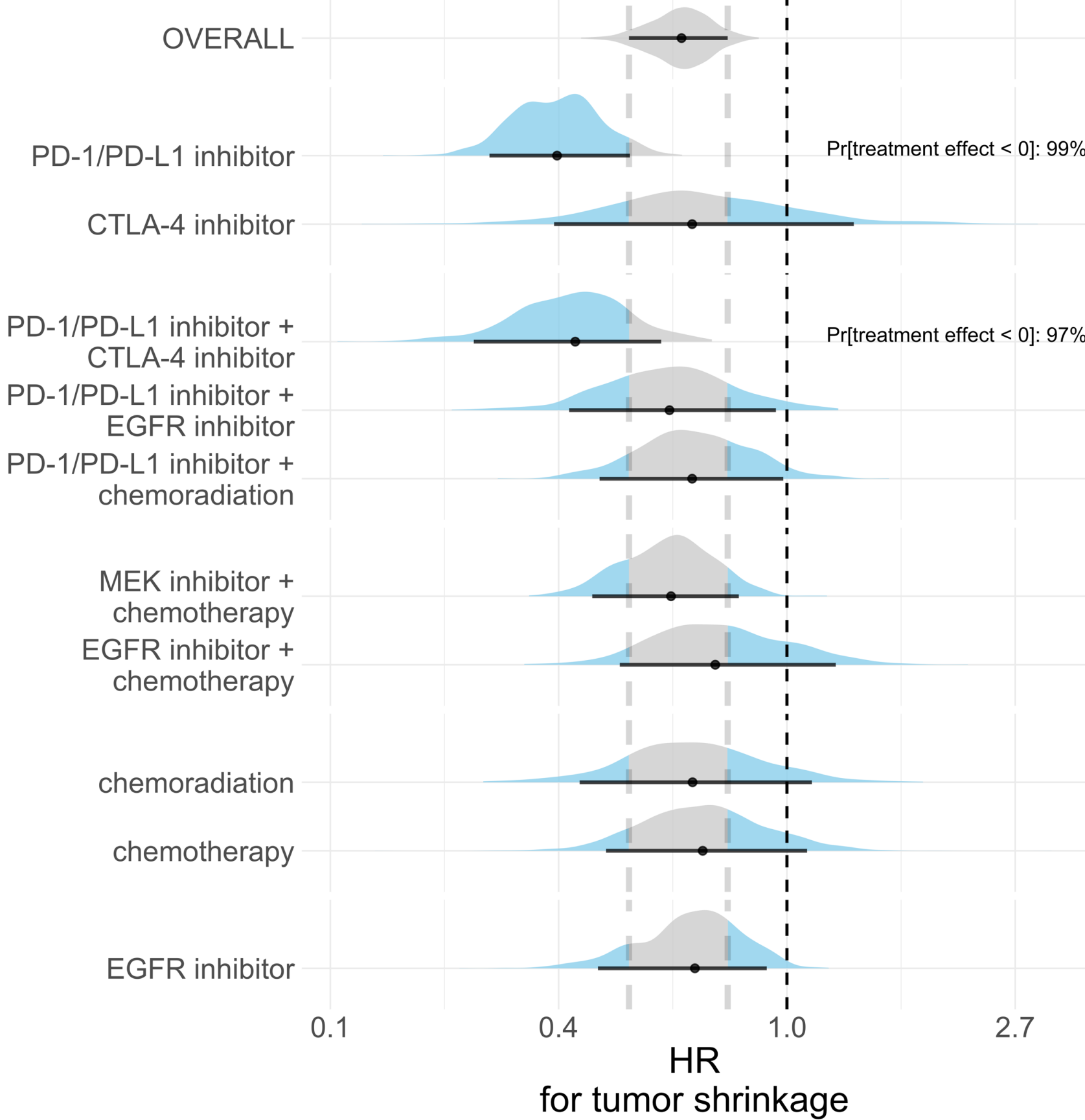
There was an increased risk of mortality among patients with higher tumor regrowth. There was negligible variation in this association by treatment.



The points represent posterior median values per treatment, with lines representing 90% posterior credible intervals (CrI). The hazard ratios are estimated hierarchically with overall mean (shown above as "OVERALL"), treatment-specific effects, and an estimated variance across treatments.

Effect of tumor shrinkage on survival

The hazard associated with tumor shrinkage varies according to the class of treatment in a joint model for tumor size and overall survival.



The hazard ratios (HRs) shown here reflect the relative hazard for mortality associated with each unit change in tumor shrinkage in a model stratified by study and cohort and adjusted for the association between tumor regrowth and mortality. Across all treatments, patients with high tumor shrinkage had better overall survival than patients with less shrinkage (median HR = 0.6).

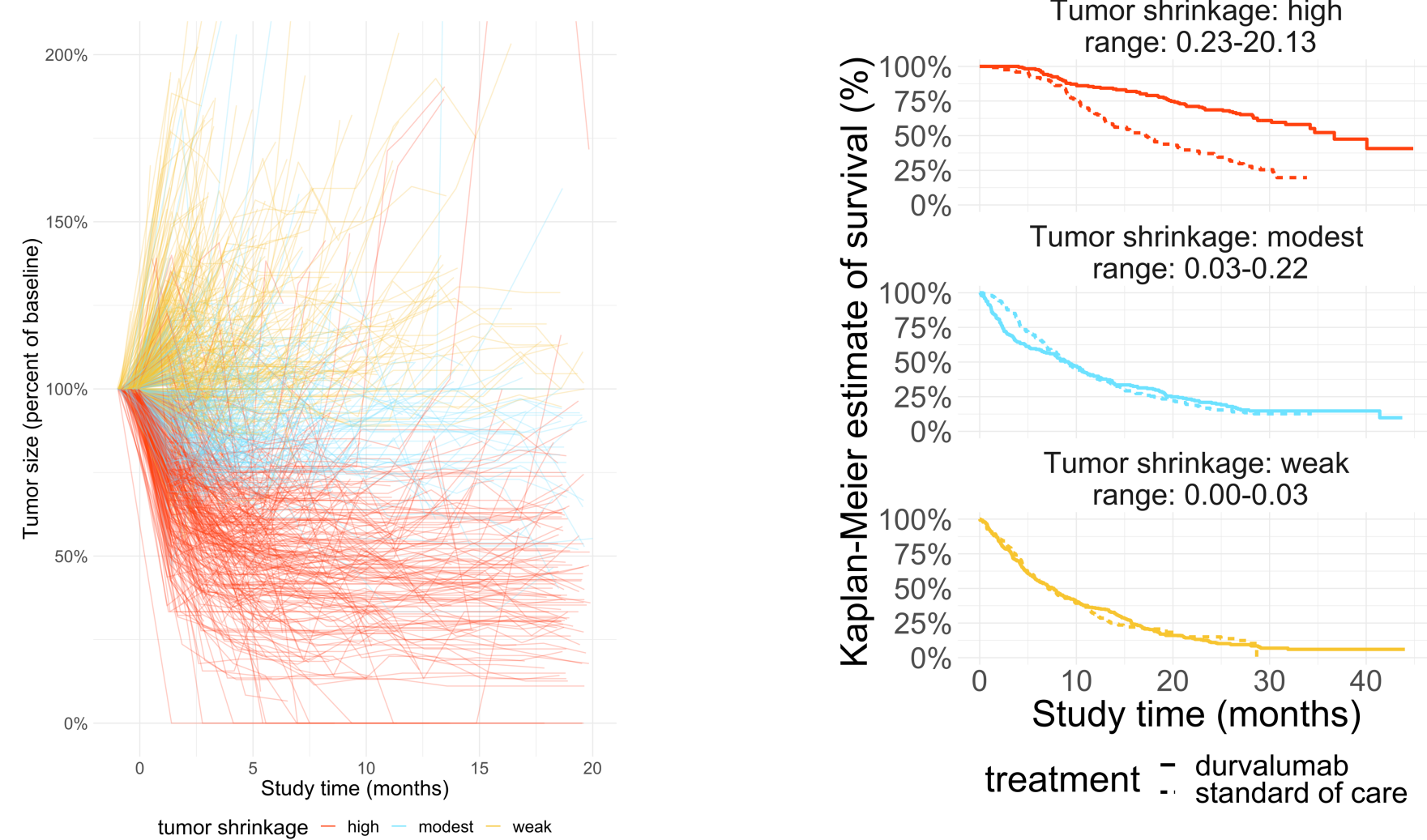
The HRs for shrinkage are estimated hierarchically with an overall mean (shown above as "OVERALL"), treatment-specific effects (shown above per treatment class), and an estimated variance among treatment effects.

Two treatments had >90% posterior probability of a treatment-specific effect ($\Pr[HR_{\text{treatment}}/HR_{\text{overall}} < 1]$), shown above as $\Pr[\text{treatment effect} < 0]$: the PD-L1 inhibitor durvalumab and its combination with the CTLA-4 inhibitor tremelimumab.

K-M by tertile of tumor shrinkage

To aid interpretation of this finding, we report the Kaplan-Meier estimate of survival according to tertiles of tumor shrinkage.

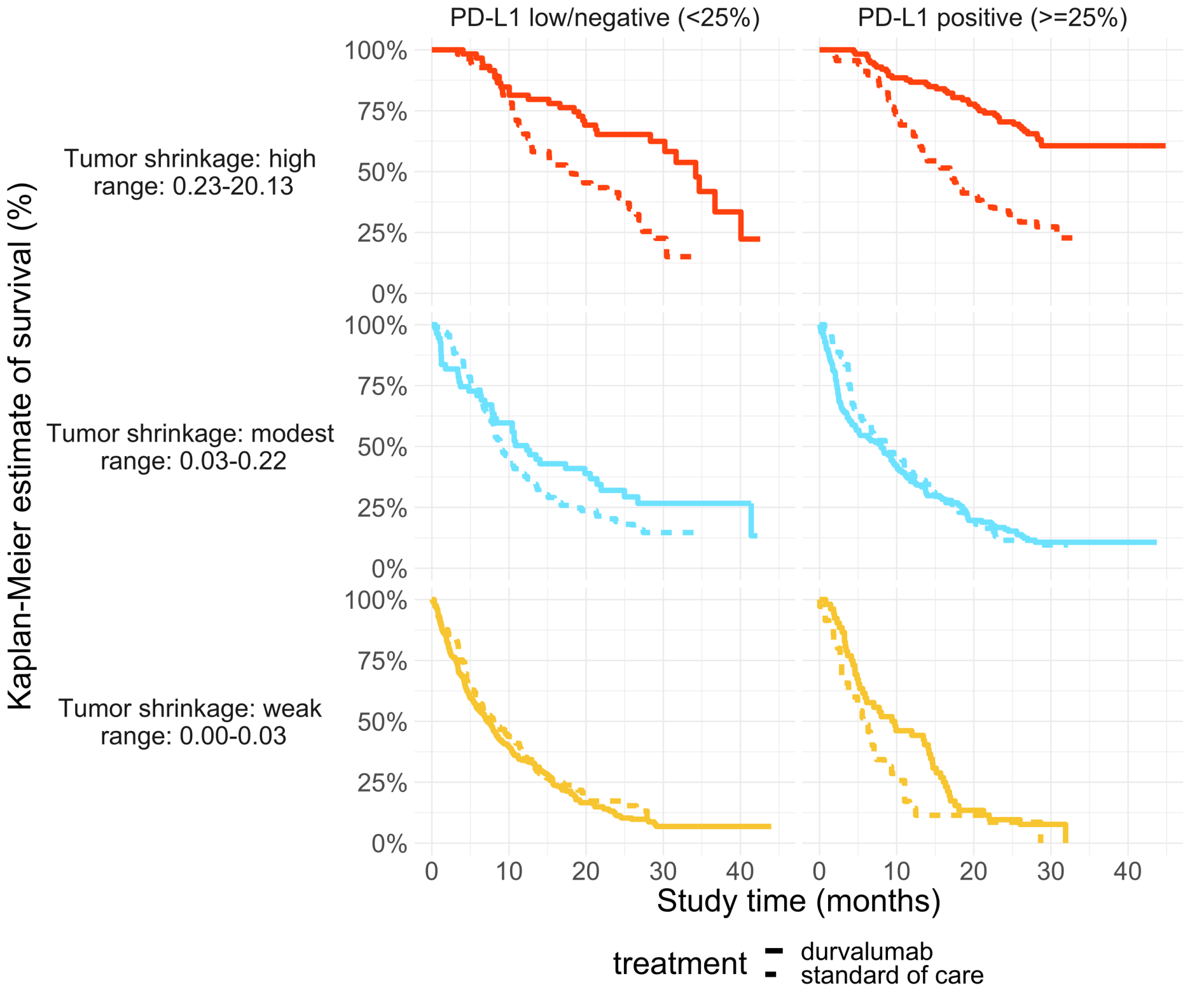
For example, among the studies including durvalumab, subjects with a similar degree of tumor shrinkage have different K-M survival probability depending on the randomized treatment regimen.



K-M by tertile of tumor shrinkage & PD-L1 expression on tumor cells

Similarly, we find evidence both in the model estimates and the observed data that this effect is independent of PD-L1 expression on tumor cells.

In an analysis by tertile of tumor shrinkage and by PD-L1 expression on tumor cells (among the studies with appropriate cohorts: Study 1108, ARCTIC, and MYSTIC), patients with high tumor shrinkage and who were randomized to receive durvalumab had better overall survival than those with high tumor shrinkage in a chemotherapy arm irrespective of PD-L1 status.



Conclusions

- Our results suggest that not all reductions in tumor size are equal. Among NSCLC patients with the same degree of tumor shrinkage, those randomized to receive a PD-1/PD-L1 inhibitor have a lower risk of mortality (better overall survival) than those receiving another regimen.
- More research is needed to determine whether our results are unique to this particular PD-L1/PD-1 inhibitor (durvalumab), and whether the trends identified transfer to other cancer types.
- The variance in prognostic value of changes in tumor size according to treatment is nonetheless interesting for several reasons. It suggests the presence of an "effect multiplier" for this immunotherapy treatment, by which the initial benefit of a reduction in tumor size leads to a survival improvement several-fold higher than is seen with other therapies. Further research is needed to evaluate whether the effect multiplication can be attributed to a durable activation of the adaptive immune system or to another mechanism.
- Finally, our results raise questions regarding the utility of surrogate endpoints based on RECIST. Such measures may be insensitive to treatment effects not mediated solely by reductions in tumor size.

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